Paleomagnetism and geochemistry of Carboniferous Sandwich Bay dykes from coastal Labrador

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Paleomagnetic, petrographic, and geochemical data are used to characterize a suite of dykes in the Sandwich Bay area, south of Cartwright, Labrador. These dykes are Carboniferous in age and their separate identity is indicated by the following: (*i*) distinct paleomagnetic signatures, (*ii*) homogeneous geochemistry, (*iii*) cross-cutting relations with other dated rock units in the area, and (*iv*) a whole-rock K−Ar date of 327 ± 13 Ma.

The dikes have a soft present Earth's field component superposed on a stable magnetization at $D = 325.5°$, $I = -26.9°$ ($N = 14$, $k = 91.6$, $\alpha_{95} = 4.6°$) with the corresponding paleopole at 16.2°N, 157.6°E (dp, dm = 2.8°, 5.2°). The stable magnetization is shown to be primary from baked contact tests. No Kiaman overprint is present. The paleopole determined for the dykes suggests emplacement of the dykes at equatorial latitudes during the Carboniferous times.

Les résultats d'une étude paléomagnétique, pétrographique et géochimique permettent de caractériser une série de dykes affleurant dans la région de Sandwich Bay, au sud de Cartwright, Labrador. Ces dykes sont d'âge carbonifère et se distinguent par (*i*) leurs signatures paléomagnétiques particulières, (*ii*) leur composition géochimique homogène, (*iii*) leurs relations de recoupement avec d'autres unités lithologiques datées dans le région et (*iv*) leur âge K−Ar sur roche totale de 327 ± 13 Ma.

Les dykes sont légèrement surimpressionnés par la composante du champ magnétique actuel tandis que leur aimantation stable est caractérisée par $D = 325{,}5°$, $I = -26{,}9°$ ($N = 14$, $k = 91{,}6$, $\alpha_{95} = 4{,}6°$), avec le paléopôle correspondant à 16,2°N, 157,6°E (dp, dm = 2,8°, 5,2°). Les tests de contact indiquent que l'aimantation stable est primaire. Il n'y a pas de surimpression due à l'inversion du Kiaman. Ce paléopôle déterminé à partir des dykes suggère leur mise en place à des latitudes équatoriales durant la période du Carbonifère.

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Objective of the study

Paleomagnetic studies and particularly the construction of apparent polar wander paths with respect to various plates have provided vital information in global tectonics (Runcorn 1956; Van der Voo 1988). The paleomagnetic signature of a rock unit, being a unique parameter, can also be employed, in conjunction with other studies, to "fingerprint" intrusives. The objective of this study was to use the paleomagnetic signature, in conjunction with regional mapping, geochemistry, and wherever possible, radiometric age determinations, to identify igneous intrusive units with the same paleomagnetic signature in the coastal Labrador portion of the Canadian Shield. Used in conjunction with radiometric age determinations, this would improve the paleomagnetic data base relative to North America.

Systematic regional mapping and geochemistry analyses have been done over the past few years on various igneous bodies intruding the coastal Labrador area (Gower *et al.* 1985, 1987). This communication, on a suite of dykes from the Sandwich Bay area, is the first of several reports documenting the paleomagnetic, field, petrographic, and geochemical characteristics of mafic intrusives from southeast Labrador.

Regional setting

The Sandwich Bay dykes (new name) are located along the southern shore of Sandwich Bay, except for one sample from Hawke Bay (Fig. 1 and insets). The dykes intrude metasedimentary and granitoid gneisses formed prior to and during the ca. 1650 Ma Labradorian orogeny and further tectonized during Grenvillian events dated at ca. 1030 or 970 Ma (Scharer and Gower 1988). Dykes from sites adjacent to White Bear River (Fig. 1, inset) intrude a coarse-grained, north-northeast-trending gabbroic dyke dated at 615 ± 2 Ma (U−Pb, zircon and baddeleyite, Kamo *et al.* 1989; Murthy *et al.* 1988). The age of the dykes is further constrained by an age of 327 ± 13 Ma (K−Ar, whole rock; Table 1), which we interpret as dating the time of instrusion. The Sandwich Bay localities considerably extend the region in eastern Canada known to have been affected by Devono-Carboniferous magmatism (Fig. 1) and also provide a geographic link between the Carboniferous basins and the minor mafic intrusions farther northwest in the Aillik Bay area. These intrusions have yielded ages (recalculated) of 353 ± 42 Ma from a diabase dyke (K−Ar, whole rock) and 338 ± 14 Ma and 405 ± 16 Ma from two lamprophyre dykes (K−Ar, phlogopite; Wanless *et al.* 1973). The ages were considered anomalously young by Wanless *et al.* but, on the basis of data reported here, can now be reappraised as possibly recording their intrusion.

The trend of the dykes (Table 2) is variable in detail but shows an overall east to east-southeast direction. No dyke has

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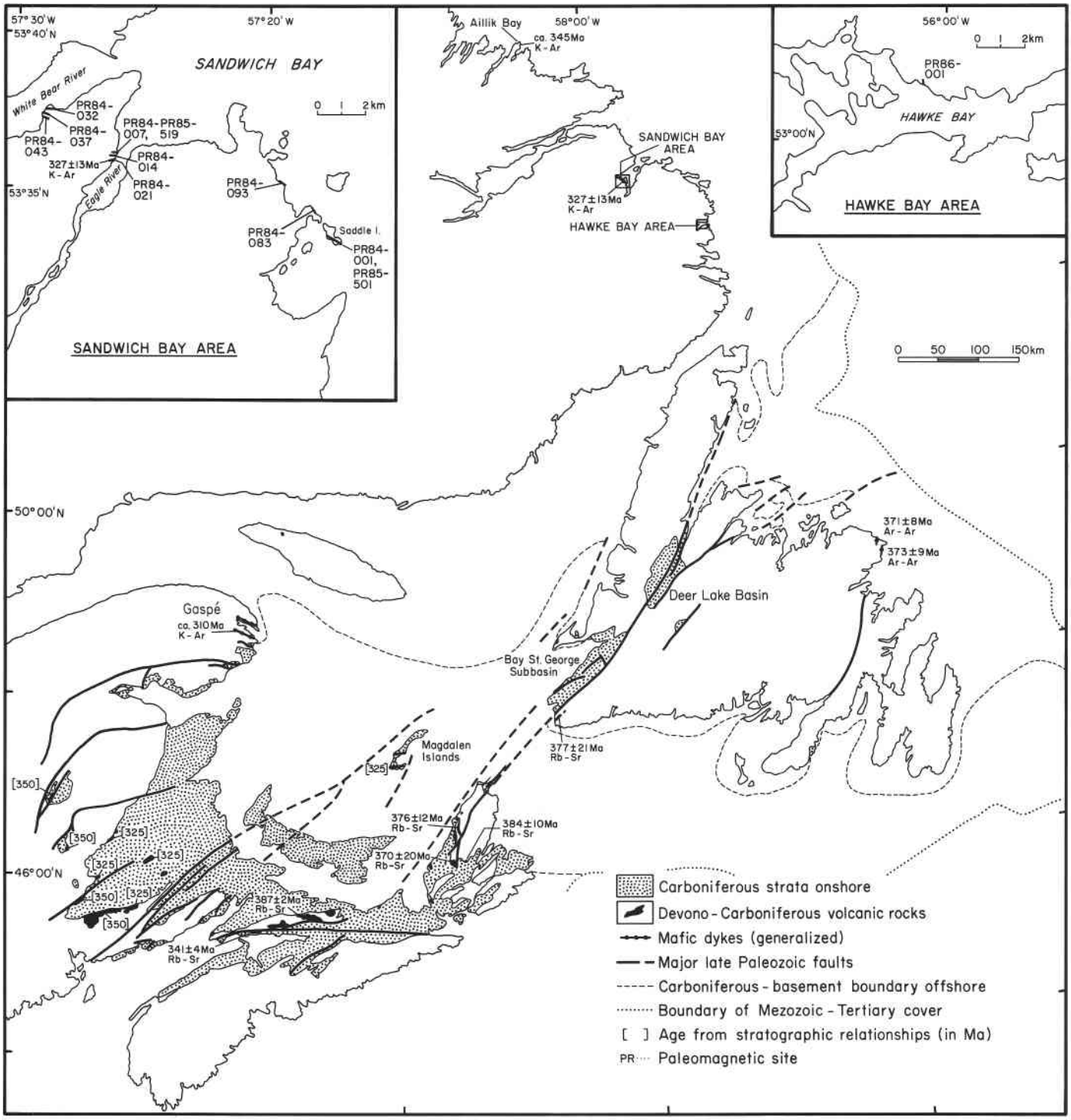


FIG. 1. Location of Sandwich Bay dykes in relation to Carboniferous basins and Devono-Carboniferous volcanic rocks in eastern Canada. Distribution of Carboniferous basins and faults is after Knight (1983), Keppie *et al.* (1982), and Bradley (1982). The distribution of Devono-Carboniferous volcanic rocks, the Rb–Sr geochronological data, and ages determined from stratigraphic relationships are from Barr *et al.* (1985), Blanchard *et al.* (1984), Dostal *et al.* (1983), Fyffe and Barr (1986), Wilton (1983), and sources cited by these authors. K–Ar and Ar–Ar data on mafic dykes are from sources reported by Seguin (1987), Murthy (1983), and Wanless *et al.* (1973).

been extrapolated beyond a single outcrop. It is unlikely that dykes regionally on strike (e.g., the Saddle Island locality and two farther northwest; Fig. 1, inset) link up directly, although they may be related to a common fracture set. The Hawke Bay locality is regionally on strike with the east-southeast-trending Sandwich Bay dykes, and the Aillik Bay dykes were also recorded as having a southeast trend (Wanless *et al.* 1973). It is noteworthy that the Gaspé dykes (Seguin 1987), which occur in a similar tectonic position relative to the Carboniferous basins, also have a similar trend.

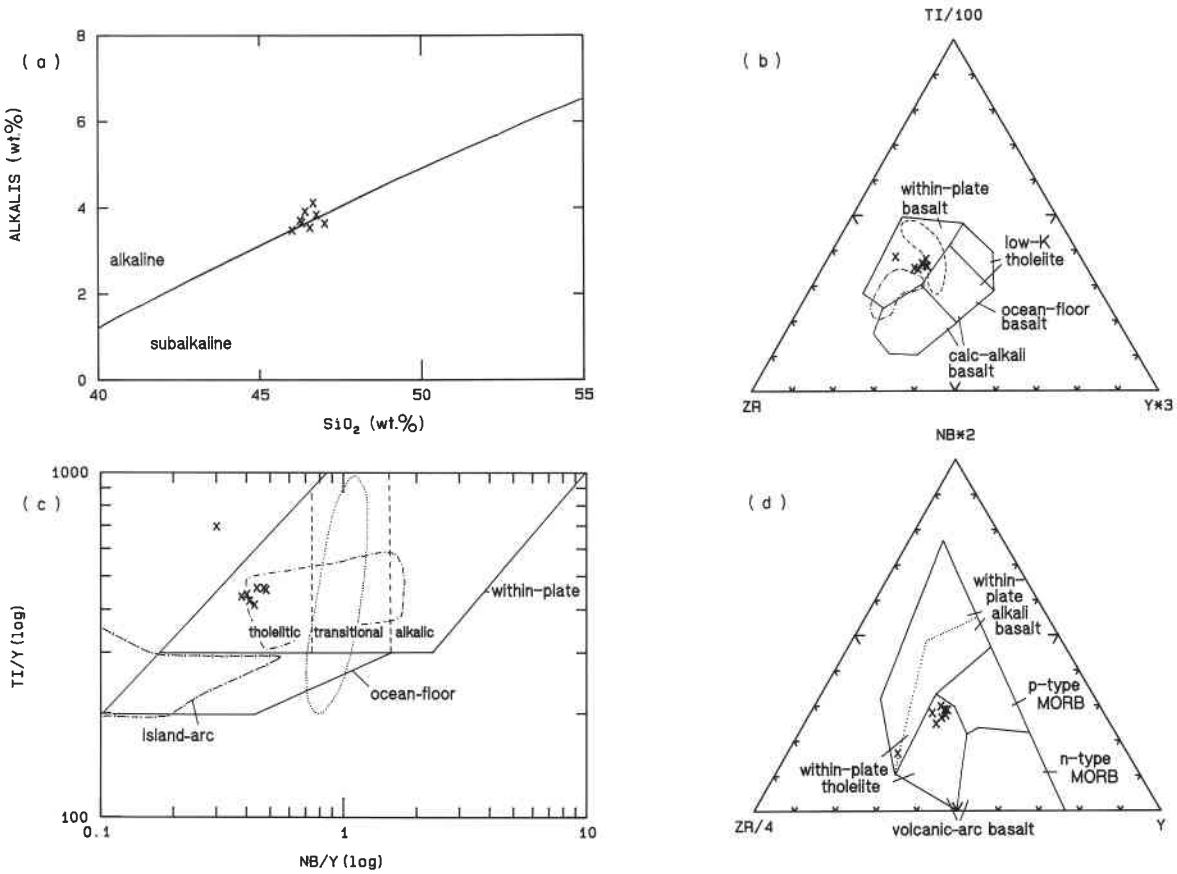


FIG. 2. Geochemical discrimination plots for the Sandwich Bay dykes (crosses); see text for discussion. (*a*) Alkalis−SiO_2 plot including the alkaline−subalkaline basalt dividing line of Irvine and Baragar (1971). (*b*) Zr−(Ti/100)−Y*3 plot after Pearce and Cann (1973); dashed field encompasses Devono-Carboniferous basalts from Cape Breton (Blanchard *et al.* 1984); dash−dot field encompasses similar-aged basalts from the Magdalen Islands (Barr *et al.* 1985). (*c*) (Ti/Y)−(Nb/Y) plot after Pearce (1982); dotted field encompasses late Visean to Namurian basalts from New Brunswick (Fyffe and Barr 1986); dash−dot field as for *b*. (*d*) (Zr/4)−(Nb*2)−Y plot after Meschede 1986).

TABLE 1. Whole-rock K−Ar data for dyke PR84-021

K (wt.%)	$^{40}Ar_{rad.}$ (ppm)	$^{40}Ar_{rad.}/^{40}Ar_{tot.}$	Age (Ma)
0.847	0.02102	0.873	327±13

NOTES: Analyst: Krueger Enterprises Incorporated, Geochron Laboratories Division. Constants used: $\lambda_\beta = 4.962 \times 10^{-10}$ year^{-1} $\lambda_\epsilon = 0.581 \times 10^{-10}$ year^{-1} $^{40}K/K = 1.193 \times 10^{-4}$ g per g K.

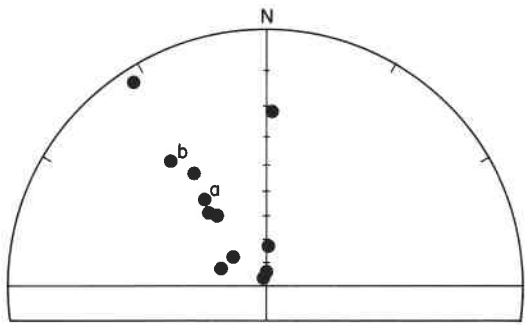


FIG. 3. Dyke mean directions at the NRM level plotted on a Wulff stereographic projection. Solid dots indicate positive inclinations; a and b refer to directions of the host rock from the baked zone (see text).

Field characteristics and mineralogy

The dykes weather black, are fine to medium grained (0.1−0.3 mm) and massive, and do not show evidence of deformation or metamorphism. They invariably contain plagioclase phenocrysts, which are up to 1 cm long. Rarely a few olivine phenocrysts, up to 0.5 cm in length, are also present. The plagioclase phenocrysts (An_{65-60}) are fresh to lightly sericitized, euhedral, and well twinned and have stubby to lath-like shapes. Interiors of grains are normally zoned, grading into an outer shell showing oscillatory zoning. The groundmass comprises olivine, plagioclase (approximately An_{60-55}), clinopyroxene, and opaque and accessory phases. Olivine occurs as euhedral to subhedral grains, partly or completely altered to a green or yellow−brown serpentine product and opaque minerals. Matrix plagioclase differs from the phenocrysts in being more elongate and commonly showing skeletal quench habit. It is fresh to lightly sericitized, well twinned, and locally zoned. Clinopyroxene (cf. titanaugite) occurs as anhedral, elongate grains having distinct purplish brown colour and weak to moderate pleochroism. It is intergrown with opaque grains that appear to be at least partly primary. The opaque mineral, probably ilmenomagnetite, is steely blue

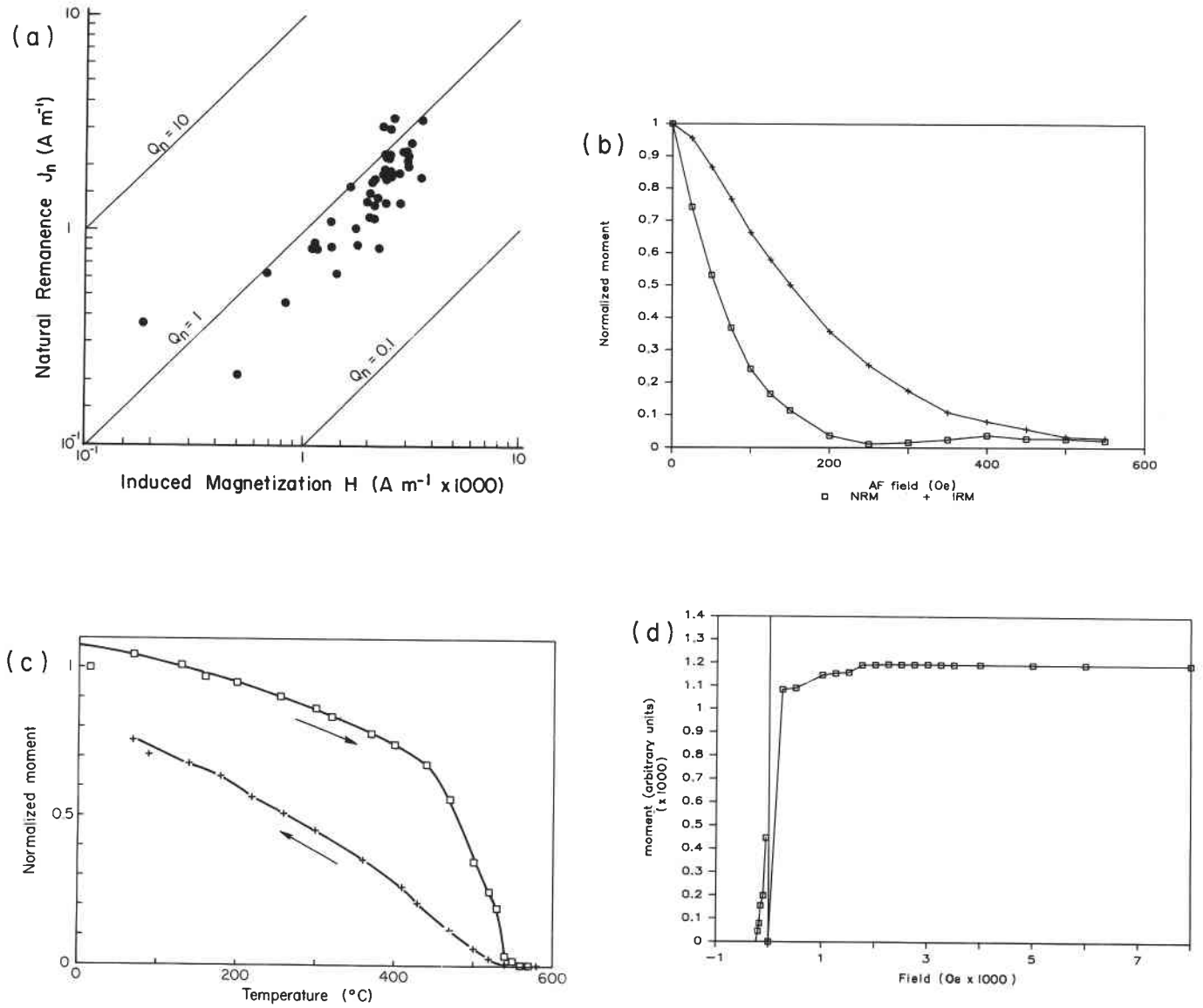


FIG. 4. (*a*) Plot of remanent magnetization. J_n, versus induced magnetization. H, showing the values of the Koenigsberger ratio, Q_n. (*b*) Comparison of AF demagnetization characteristics of NRM and IRM (Lowrie–Fuller test) for sample PR84-005A to determine the domain state of the remanence carriers. On the horizontal scale, 600 Oe = 60 mT. (*c*) Plot of normalized magnetic moment versus temperature (J_s–T) for sample PR84-034 in a field of 8600 Oe (860 mT), heating and cooling. (*d*) IRM acquisition and back-field characteristics for sample PR84-032B. On the horizontal scale, 1000 Oe = 100 mT.

in reflected light and occurs as equant to elongate grains. It is most abundant in the matrix, but some larger grains are present. Accessory phases include a dark-brown phase (cf. perovskite) intergrown with the opaque mineral and needles of skeletal apatite. Textural relationships in the groundmass clearly indicate that clinopyroxene crystallization postdated that of plagioclase and olivine and suggest that plagioclase postdated olivine.

Geochemistry

Of the 12 sites investigated for their paleomagnetic signature, two are in country rocks (PR85-501 and PR85-519) and two (PR84-014 and PR84-021) are in an outcrop at which three dykes are present. Geochemical data for the third dyke (PR84-007) in this outcrop and for the seven remaining sites are reported in Table 2. Major elements were analyzed by atomic absorption spectroscopy (AAS), and trace elements were analyzed either by AAS (Cr, Ni, Cu, Zn, Rb, Sr, Mo, Ba, Pb) and (or) by X-ray fluorescence (V, Cr, Ga, Rb, Sr, Y, Zr, Nb, Ba, La, Ce, Th), except for F and U, which were analyzed by selective-ion electrode and neutron activation methods, respectively.

Of particular significance is the compositional homogeneity of the dykes (Table 2). This is evidenced by the tight clustering of points in the variation diagrams (Fig. 2), the sample from Hawke Bay being a partial exception. There is no reason to suspect that the differences in the Hawke Bay sample are analytical. The differences in chemistry, restricted within analytical precision to higher TiO_2 and Zr and lower Y and Nb of the Hawke Bay dyke, are empirically correlated with its geographical isolation from the other samples. The dyke nevertheless shows a paleomagnetic signature identical to those of the

TABLE 2. Field, geochemical, and paleomagnetic data

	Mafic					
Dyke:	PR84-001	PR84-007	PR84-014	PR84-021	PR84-032	PR84-037
Samples:	PR84-001 – PR84-006	PR84-007 – PR84-013	PR84-014 – PR84-020	PR84-021 – PR84-024	PR84-032 – PR84-036	PR84-037 – PR84-042
						PALEOMAGNETIC
n/N	11/11	4/4	5/5	6/6	10/10	10/10
						Natural remanent
D (°)	312.4	2.0	292.4	327.7	348.3	322.5
I (°)	70.8	21.3	68.9	35.0	86.4	50.5
α_{95} (°)	15.0	52.8	17.1	13.2	18.4	18.3
k	10	4.0	21	27	7.8	8.0
						Characteristic
D (°)	329.1	331.9	312.5	324.7	327.4	322.6
I (°)	−28.1	−33.4	−14.8	−36.0	−27.4	−26.1
α_{95}	7.5	3.8	28.5	9.9	8.5	8.6
k	24	363	8.4	24	18	18
						GEOCHEMICAL
SiO_2 (%)	46.00	47.00			46.25	46.75
TiO_2	3.43	3.32			3.51	3.32
Al_2O_3	13.60	13.75			14.20	14.60
Fe_2O_3	3.00	3.82			3.60	3.37
FeO	11.46	10.92			11.33	11.26
MnO	0.28	0.23			0.23	0.22
MgO	5.68	5.40			5.48	5.87
CaO	9.07	9.20			9.02	9.14
Na_2O	2.76	2.65			2.78	2.79
K_2O	0.71	0.98			0.92	1.04
P_2O_5	0.68	0.66			0.68	0.56
LOI	2.00	1.18			0.55	0.73
Total	98.67	99.11			98.55	99.65
F (ppm)	505	415			505	429
V	374	335			343	338
Cr	50	38			81	94
Ni	30	31			34	38
Cu	49	43			45	46
Zn	128	126			139	131
Ga	24	22			20	25
Rb	14	19			24	20
Sr	328	343			355	418
Y	47	43			46	43
Zr	247	241			230	203
Nb	18	20			22	19
Mo	4	4			5	4
Ba	537	548			520	488
La	21	27			10	7
Ce	65	57			44	25
Pb	1	1			1	1
Th	1	1			1	1
U	0.5	0.6			0.5	0.4
						FIELD
Trend of dyke (°)	120	090	088	077	095	108
Width of dyke (m)	15	0.3	1.3	0.3	1.0	1.2

NOTES: Dykes are numbered the same as that of the first sample number it contains. Thus, dyke PR84-001, for example includes samples PR84-001 to PR84-006 and so on; *n*, number of specimens yielding characteristic magnetization; *N*, number of specimens studied; *D* and *I*, declination and inclination of the mean direction; *k* and α_{95}, Fisher's precision parameter and the radius of the 95% circle of confidence; LOI, loss on ignition. Pb and Th are at detection limit; blanks indicate that analytical data are unavailable.

for Carboniferous dykes from coastal Labrador

dykes				Contact tests	
PR84-043 PR84-043– PR84-050	PR84-083 PR84-083– PR84-092	PR84-093 PR84-093– PR84-097	PR86-001 PR86-001– PR86-007	PR85-501 PR85-501– PR85-518	PR85-519 PR85-519– PR85-527
DATA					
8/8	11/11	4/5	5/5	9/9	9/9
magnetization					
325.5	5.1	4.7	327.2	324.9	322.9
53.3	83.9	72.3	3.2	45.4	27.2
19.8	14.3	26.1	233	33.4	67.9
8.8	11.1	9.5	1.4	3.3	4.4
magnetization					
325.9	227.0	323.5	334.8	323.5	325.7
−35.3	−50.8	−21.5	−19.1	−24.0	−28.7
10.8	16.9	14.6	7.7	6.5	6.8
16	16.0	24.0	47.0	42.0	224.0
DATA					
46.65	46.55	46.40	46.30		
3.52	3.48	3.47	3.84		
14.19	14.04	14.21	13.78		
4.00	4.26	3.61	4.67		
10.95	10.86	11.45	10.44		
0.23	0.24	0.23	0.24		
5.54	5.52	5.49	5.82		
8.70	9.39	9.02	9.24		
2.86	2.73	2.79	2.72		
1.25	0.80	1.12	0.93		
0.68	0.66	0.67	0.67		
0.90	0.55	0.57	0.96		
99.47	99.08	99.03	99.61		
543	517	555	475		
348	341	338	357		
83	80	75	87		
34	34	33	36		
47	43	45	45		
142	145	137	141		
26	25	27	31		
34	19	22	14		
415	429	380	399		
51	47	49	33		
232	231	228	274		
22	19	20	10		
5	5	4	5		
617	560	545	571		
10	8	13	21		
35	35	55	67		
1	1	1	1		
1	1	1	1		
0.5	0.4	0.4	0.4		
DATA					
108	150	130	170		
5.5	0.3	12.0	0.3		

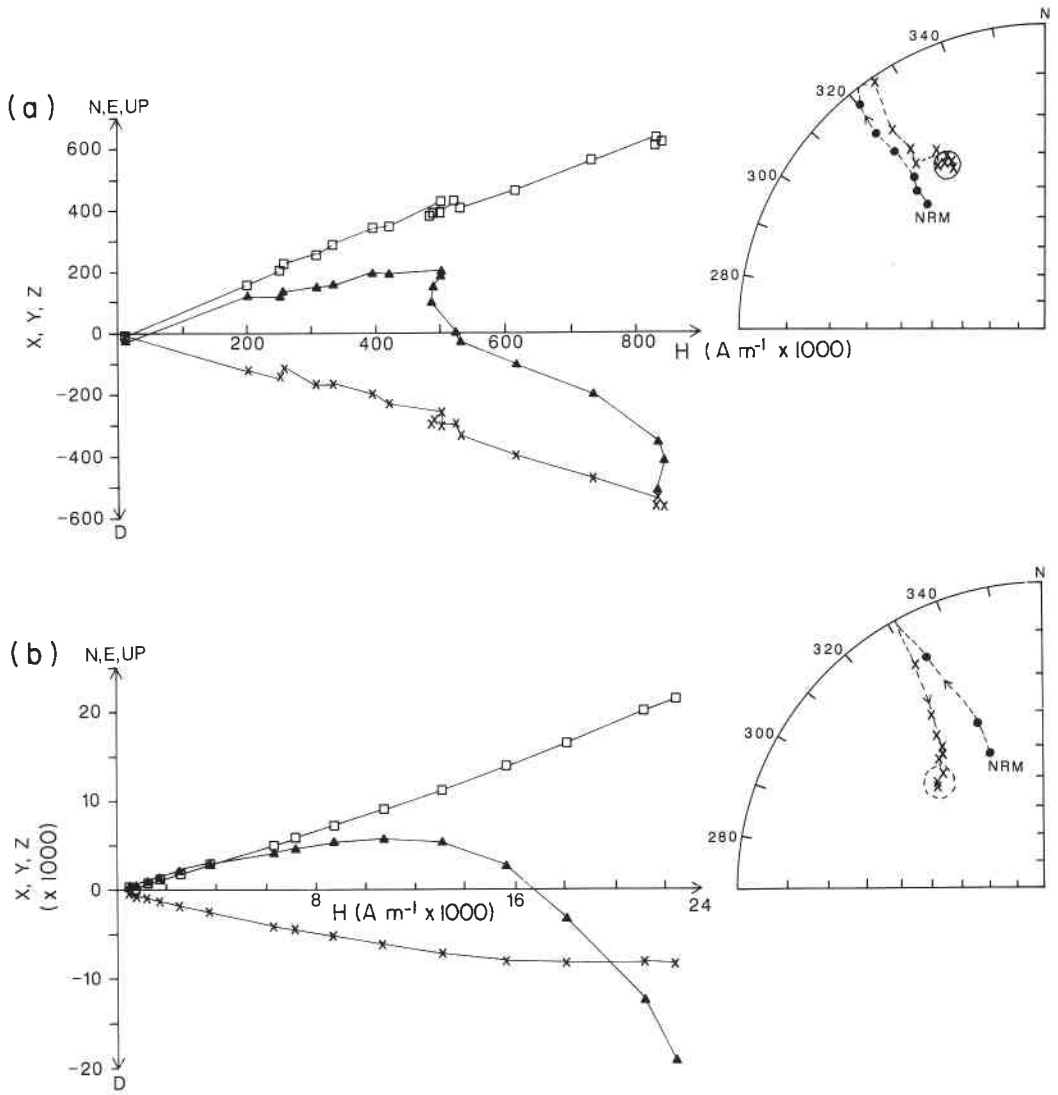


FIG. 5. (*a*) Response of specimens PR84-002A (dyke PR84-001) to AF (20 mT) and thermal (570°C) demagnetizations. (*b*) Response of specimen PR84-022A (dyke PR84-021) to AF (50 mT) demagnetization. In the vector diagrams on the left, variations of *X*, *Y*, *Z* are plotted versus *H*, the horizontal component. The units are arbitrary. Squares: *X* vs *H*; Crosses: *Y* vs. *H*; Triangles: *Z* vs. *H*. In the stereographic projections (Wulffs net) on the right, solid dots indicate positive inclinations and crosses indicate negative inclinations.

dykes from Sandwich Bay (see below).

The lack of chemical variability in the dyke components, including those components generally considered mobile, supports the petrographic observation that these rocks are only slightly altered if at all, a conclusion further indicated by their low oxidation state and low loss-on-ignition values.

The rocks are best considered olivine tholeiite having some features of alkali basalt and, from a tectonic standpoint, are classified as within-plate tholeiites (Figs. 2*a*–2*d*). Included for comparative purposes are the outlines of fields for mafic Devono-Carboniferous volcanic rocks from Nova Scotia and late Visean to Namurian (ca. 325 Ma) basaltic rocks from New Brunswick and the Magdalen Islands (data sources are quoted in the caption to Fig. 2). The data show much more scatter and are apparently more alkaline than the Sandwich Bay dykes.

A final note on the chemistry concerns dyke PR84-083. This dyke shows a paleomagnetic signature distinct from those of the other Sandwich Bay dykes, but it is clearly part of the suite on chemical criteria. The dyke is close to a major brittle fault. The difference in paleomagnetic character, despite the demonstrable chemical identity, compels us to omit this dyke from discussion in paleomagnetism. (As an aside, the chemically distinct dyke PR86-001 and the paleomagnetically distinct dyke PR84-083 further demonstrate the value of combined paleomagnetic and geochemical studies on mafic dykes.)

Paleomagnetism

At least five independently oriented samples were collected from each dyke (Fig. 1). The dykes from the Sandwich Bay area were collected during the 1984 field season. During 1985, having preliminary knowledge of the paleomagnetic behaviour of the samples collected in 1984, we collected samples of the host rock in order to determine the nature of remanence of this suite of dykes. The samples are in the form of oriented blocks or cores drilled in the field using a portable gasoline-powered field drill. The orientation was done by both solar and Silva compasses, which showed very good agreement. Two cylinders

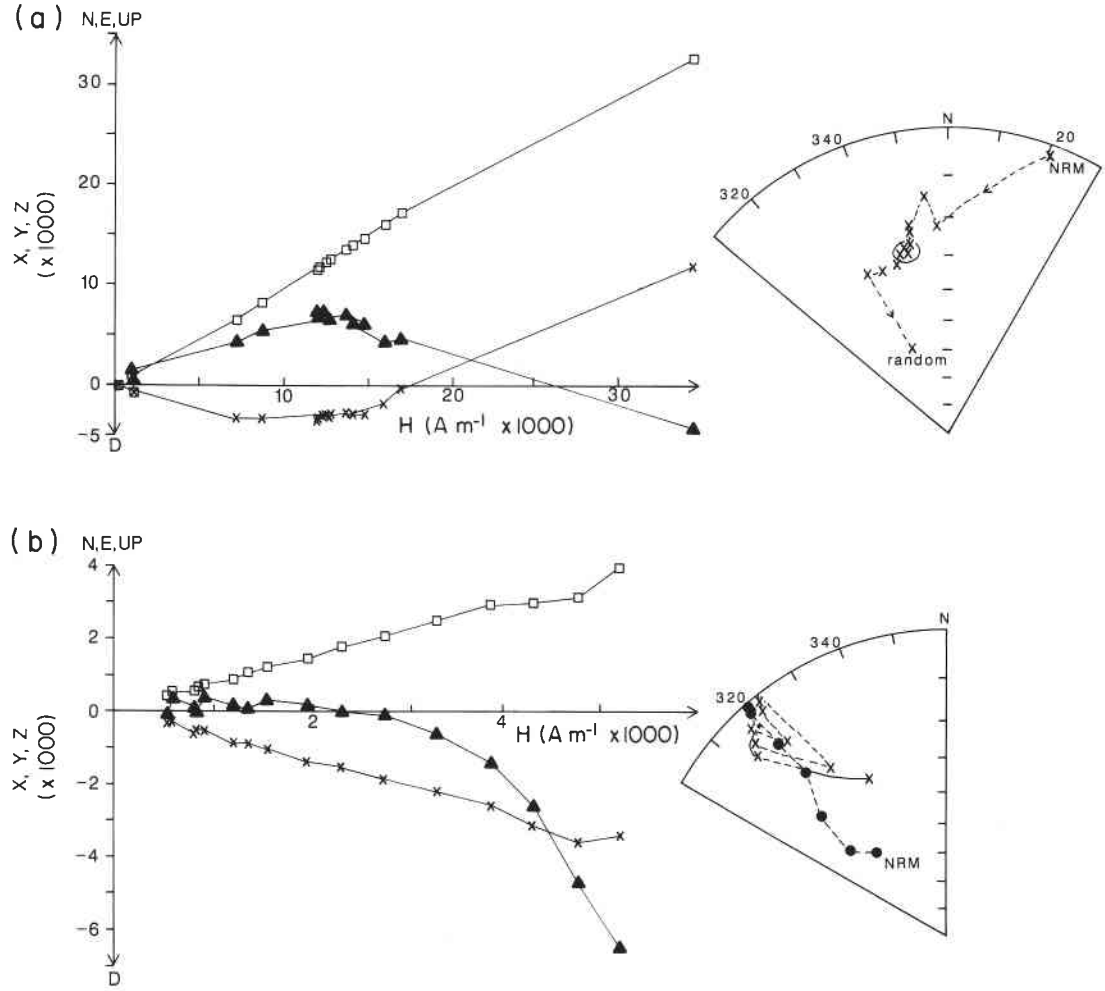


FIG. 6. Response of specimens PR84-009A (dyke PR84-007) to AF (60 mT) and thermal (570°C) demagnetizations. (*b*) Response of sample PR84-093A (dyke PR84-093) to AF (40 mT) demagnetization. Symbol meanings are the same as in Fig. 5.

(specimens) were cut from each core. Systematic sampling of the host rock at regular distances away from the dyke contact was made at various dyke locations. The host rock (banded gneiss) was also collected far from one of the dykes.

A Schonstedt SSM-1A spinner magnetometer was used for remanence measurements. The specimens were treated by alternating-field (AF) and thermal demagnetization techniques with Schonstedt demagnetizers (models GSD-1 and TSD-1, respectively). During demagnetization, the ambient field at the location of the sample was of the order of 10 nT during AF treatment and <5 nT during thermal treatment. *Each* specimen was demagnetized at very close intervals during both AF and thermal treatments (usually in about 20 or more steps), and the various components of magnetizations present were isolated for each specimen *individually*. Thermal demagnetization of the specimens was not always successful, as the remanence carrier(s) tended to become chemically altered at temperature around 500°C. Analytical technique used in recognizing various magnetizations included visual inspection of the Zijderveld diagrams (Zijderveld 1975) for straight-line segments, plotting stereonets for recognizing stable end points, vector difference (Hoffman and Day 1978) techniques, and the principal-component analyses (Kirschvink 1980). Rock magnetic studies were conducted on the specimens to investigate the nature of the magnetic mineral and its possible magnetic domain state. These included (*i*) Curie-point determinations with a vertical Curie balance (Deutsch *et al.* 1971), (*ii*) the comparison of the AF demagnetization characteristics for the natural remanent magnetization (NRM) and the isothermal remanent magnetization (IRM) for the Lowrie–Fuller test (Lowrie and Fuller 1971), and (*iii*) the IRM acquisition and the back-field experiments with a Varian Associates electromagnet capable of producing a 0.8 T field.

The dyke NRM mean directions (Fig. 3) string along a northwest–southeast great circle, having predominantly steep inclinations. Such inclinations suggest possible dominance of the present Earth's field (PEF) component. The intensity of natural remanence is of the order of 1.0 A/m. Values for Koenigsberger ratio (Fig. 4*a*) tightly cluster at slightly less than 1, indicating that the PEF component is significant. The AF demagnetization characteristics of the NRM and the IRM (Fig. 4*b*) were compared to determine the domain state of the remanence carrier (Lowrie and Fuller 1971). The IRM is harder to demagnetize than the NRM, suggesting multidomain remanence carriers, which is consistent with a significant PEF component being present. Curie-point determinations (Fig. 4*c*) are consistent with pure magnetite as the magnetic mineral, possibly becoming chemically altered during heating. IRM-

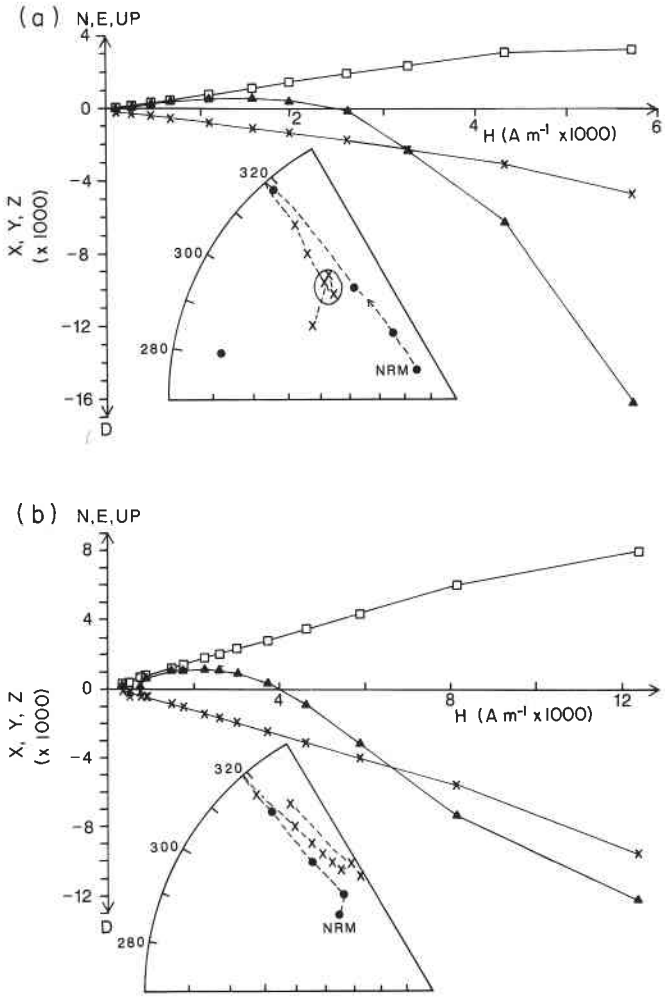


FIG. 7. Response of specimens PR84-033B (dyke PR84-032) to AF (60 mT) demagnetization. (*b*) Response of specimen PR84-045B (dyke PR84-043) to AF (65 mT) demagnetization. Symbol meanings are the same as in Fig. 5. The stereonet plot in (*a*) shows an isolated downward direction with declination around 280°, which is the last step in the demagnetization procedure. This point was not joined to the rest of the demagnetization trend for the purpose of clarity.

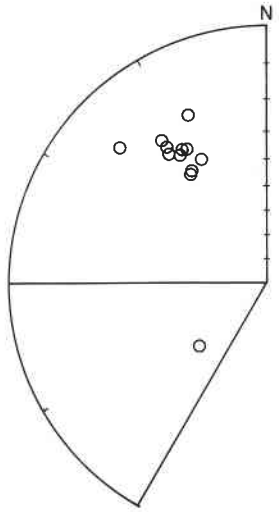


FIG. 8. Individual dyke mean directions plotted on a Wulff stereographic projection. Open symbols indicate negative inclinations. The isolated mean direction with southwest declination is for dyke PR84-083. See text for discussion.

acquisition experiments (Fig. 4*d*) show that the sample gets magnetically saturated in fields of 1500 Oe (0.15 T), also indicating that magnetite is the magnetic mineral.

Response to demagnetization is shown in Figs. 5–7. The vector diagrams are in the form of H versus X, Y, and Z rather than the conventional Zijderveld diagrams. Murthy (1985) has demonstrated the advantages of this type of representation over the conventional diagrams. After an inital removal of PEF component, a stable magnetization with northwest declination and upward inclination was recognized (Fig. 5). For specimen PR84-022A, the first few steps of demagnetization removed the PEF component, followed by simultaneous removal of two components during intermediate demagnetization steps. During at least the last eight steps of demagnetization, there seems to be only one vector being erased, as evidenced by a straight, origin-seeking segment in the vector diagram and a stable end point in the stereo plot. The specimen PR84-002A was partly AF demagnetized and was then thermally demagnetized up to 570°C. This specimen seems to carry two discrete magnetizations: a PEF component carried by the lower coercivity–blocking-temperature grain-size spectrum, followed by a characteristic magnetization retained by the higher coercivity – blocking-temperature grains. Similar responses were noted from other dykes (Figs. 6 and 7). The straight-line segments in the vector diagrams all lead to the origin, suggesting no other unrecovered component. Stable end points were also noticed in the stereonet plots.

Thermal demagnetization, in general, did not produce as clean an isolation of characteristic remanence as the AF treatment. The specimen directions developed a tendency to move erratically when the specimens were heated to beyond 500°C, probably due to the chemical alteration taking place in the magnetic minerals, with the newly formed minerals becoming magnetized in the specimen's internal field. Chemical alteration of magnetic minerals during heating was noticed during the Curie-point determinations as well (Fig. 4). Thus, except for occasional thermal treatment and occasional AF + thermal treatment, most of the specimens were treated by the AF method at very close steps. This procedure yielded clean demagnetization diagrams, straight-line segments over many demagnetization steps, and stable end points. In addition to the Sandwich Bay dykes, the dyke from the Hawke Bay exhibited a similar response, viz, a PEF component superposed over a northwest and upward stable remanence.

Results for dyke PR84-083 are also presented in Table 2. The paleomagnetic signature of this dyke is quite distinct. This dyke has a mean direction at $D = 227.0°$ and $I = -51°$, isolated using converging remagnetization circles. The PEF component is virtually absent. Although the paleomagnetic signature is quite distinct, chemically this dyke is identical to the other Sandwich Bay dykes except that it is adjacent to a major brittle fault (see discussion in earlier section). We do not yet have a satisfactory interpretation of this dyke's remanence. This distinct magnetization cannot possibly be the result of tilting alone of the dyke due to movement on the fault, as the paleomagnetic difference is too large. Because of its contrasting paleomagnetic behavior, this dyke is not included in the further statistical analyses of the paleomagnetic data for this suite. It is hoped that, in a later publication, this dyke's

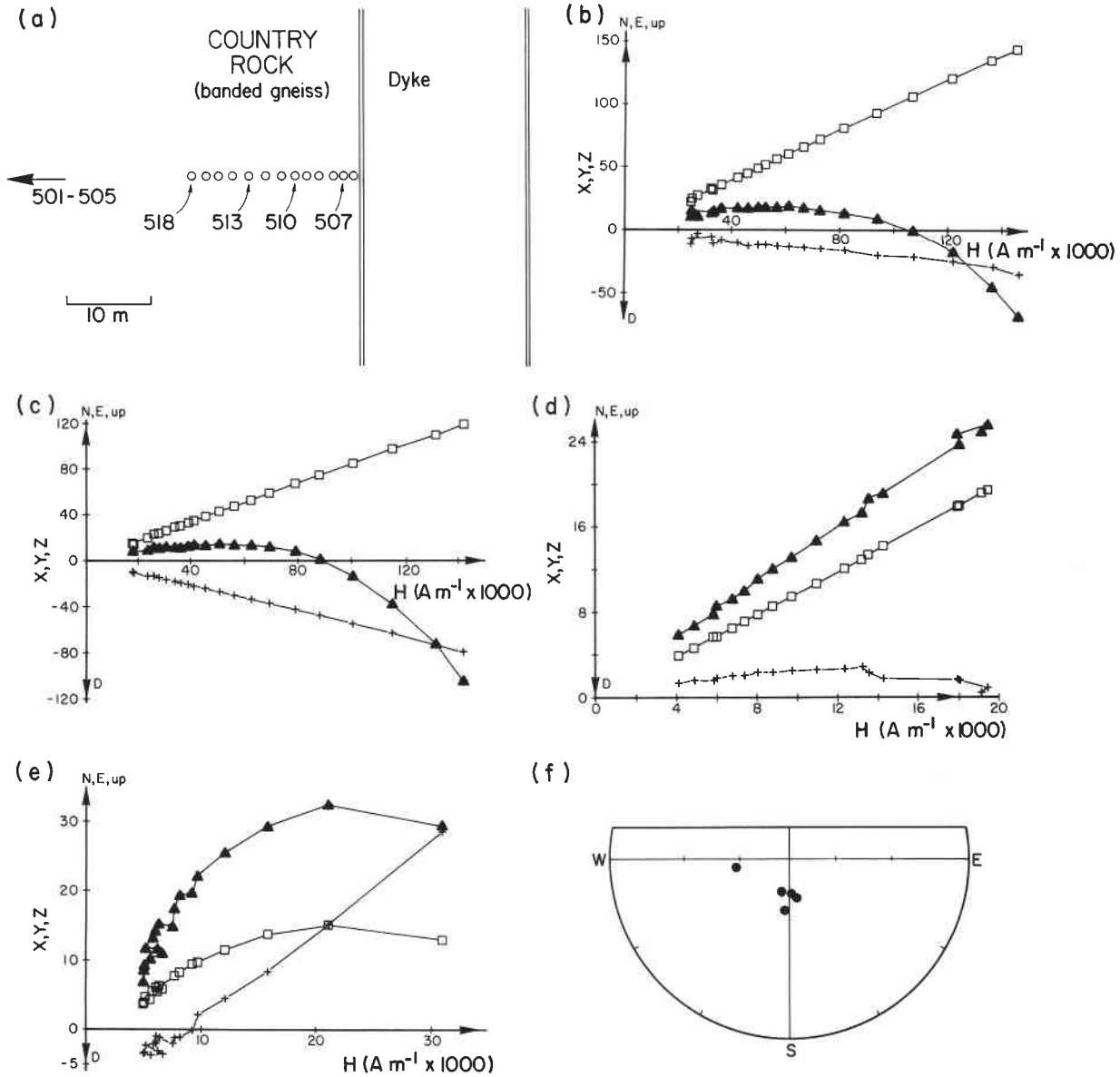


FIG. 9. Baked-contact test for dyke PR84-001. (*a*) Dyke width is shown to scale with respect to locations of host-rock samples PR85-507 to PR85-518 from within the contact zone. Samples PR85-501 to PR85-505 are from the host rock well away from the dyke contact (a few hundred metres) and are presumed to be unaffected by the dyke intrusion. (*b*), (*c*), (*d*), and (*e*) Response of baked-contact samples PR85-507, PR85-510, PR85-513, and PR85-517A, respectively, to AF demagnetization. Variations of *X*, *Y*, and *Z* components are plotted against the horizontal component, *H*. Units of vertical axis are to be multiplied by 1000. The units are arbitrary. Symbols in the vector diagrams have the same meanings as in Fig. 5. In (*f*), the magnetizations of samples PR85-501 to PR85-505 (samples away from the contact) are plotted after AF demagnetization. The grouping of this set of samples being distinct from the dyke's stable remanence coupled with contact-zone samples having similar magnetization as the dyke's suggests that the dyke's magnetization is primary (see text).

paleomagnetic signature can be more fully interpreted.

In addition to the study of vector diagrams and stereoplots, quantitative analysis by the principal-component technique (Kirschvink 1980) was made of each specimen's data. The coercivity spectra for the two recognized components of magnetizations are different, and the PCA procedures yield useful results, not different from the visual analyses. Vector-subtraction procedure (Roy and Park 1974) yielded the same result. However, the great-circle analyses (Halls 1976) were not found to be useful (except for dyke PR84-083), as all the great circles have the same northwest–southeast axes. The mean direction for each dyke was calculated from the isolated characteristic remanence for each specimen (Table 2 and Fig. 8).

Contact tests

Baked-contact tests were performed on the dykes, and the results for two of the dykes are shown in Figs. 9 and 10. For dyke PR84-001 (Fig. 9), samples of the host rock were collected extending away from the dyke margin at very close intervals. Samples PR85-501 to PR85-505 are host-rock samples a few hundred metres away from the dyke, where presumably the dyke intrusion would have no effect on the

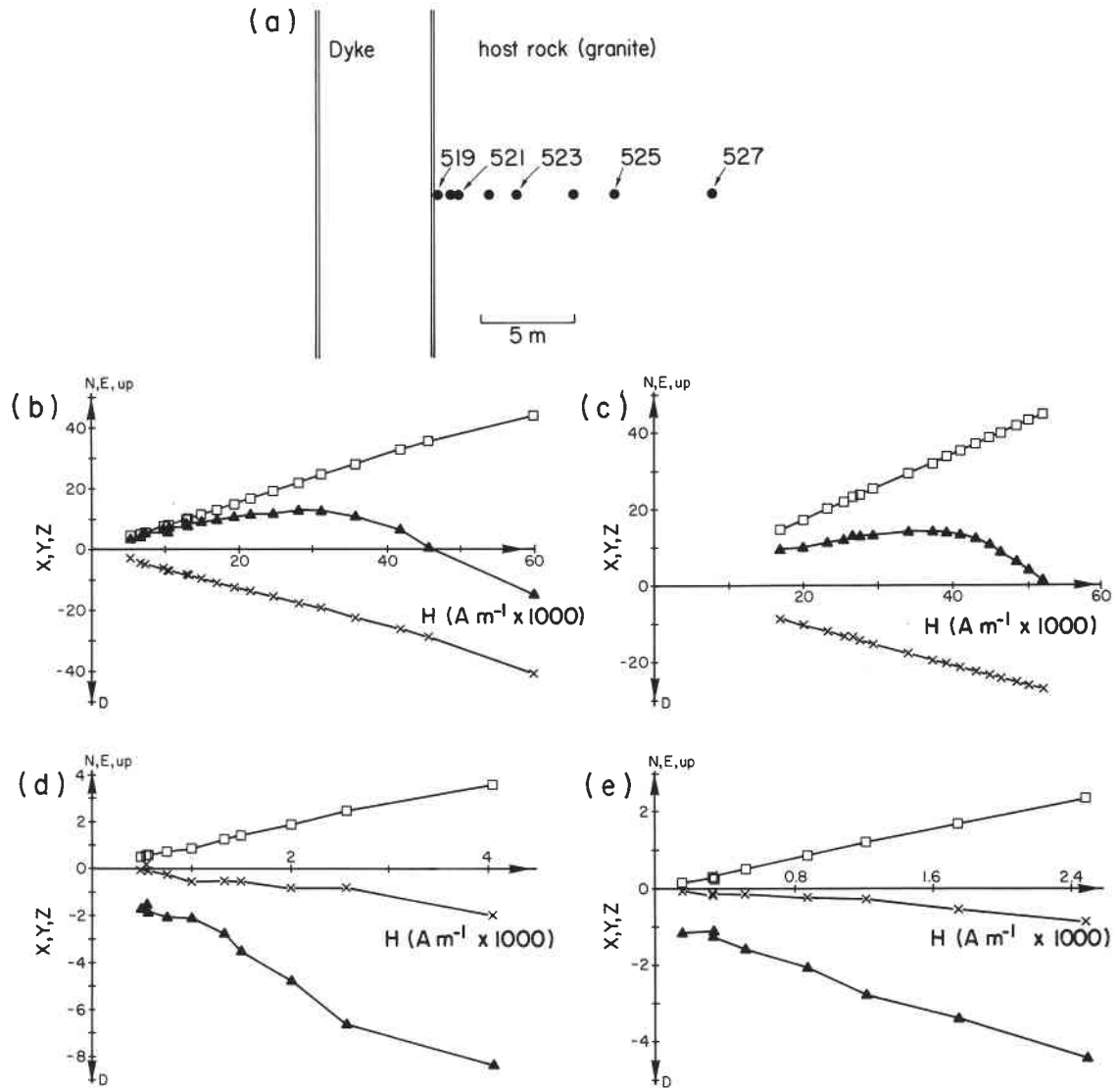


FIG. 10. Baked-contact test for dyke PR84-021. (*a*) Dyke width is shown to scale with respect to locations of host-rock samples (PR85-519 to PR85-527) from within the contact zone. (*b*), (*c*), (*d*), and (*e*) Response of specimens PR85-519B, PR85-521, PR85-525A, and PR85-527, respectively, to demagnetization plotted in the form of vector diagrams. Units of vertical axis are to be multiplied by 1000. Symbols have the same meaning as in Fig. 5.

magnetization of the host rock. Response to demagnetization of samples close to the dyke is shown in the rest of Fig. 9 in the form of vector diagrams. Samples PR85-507 and PR85-510 showed responses very similar to those of the dyke samples, indicating that the magnetization of these samples was probably reset during the heating associated with dyke intrusion. Samples PR85-513 and PR85-517 showed steeper inclinations and a somewhat erratic response. This magnetization is probably the composite of a magnetization acquired at the time of dyke injection and the original magnetization of the host rock. Well away from the dyke, the host rock showed steep positive inclinations with southerly declinations. A similar result was obtained for dyke 03 (Fig. 10). Magnetizations of samples PR85-519B and PR85-521 are similar to those of the dyke samples and were obviously reset during the time of the dyke intrusion. Samples PR85-525A and PR85-527 show soft magnetization, probably the PEF effect. The contact studies show that the dyke intrusion obviously caused resetting of magnetizations in its immediate vicinity and suggests that the stable magnetization of the dykes is most probably primary and was acquired at the time of intrusion of the dykes.

Discussion

Individual dyke directions and associated statistics are presented in Table 2; the mean and the paleopole are presented in Table 3. The improvement in both within-dyke and between-dyke grouping after partial demagnetization, the positive baked-contact tests, and the absence of a Kiaman overprint prevalent in many early and middle Paleozoic rock units all suggest that the stable direction is primary and was acquired at the time of intrusion of the dykes, probably at around 327 Ma ago. The cooling history of dykes is such that the paleosecular variation of the Earth's magnetic field is probably averaged for a number of dykes. Figure 11 shows the location of the paleopole for Sandwich Bay dykes along with other reliable Early to Middle Carboniferous paleopoles relative to North America.

TABLE 3. Mean paleomagnetic direction for the Carboniferous dykes from coastal Labrador

	N	D (°)	I (°)	α_{95} (°)	k
NRM	11	329.9	50.1	16.4	8.8
Characteristic direction	11	325.5	−26.9	4.8	91.2
Paleopole	16.2°N, 157.6°E with dp = 2.8°, dm = 5.2°				
Paleolatitude	15°S ± 3°				

NOTES: N is number of dykes; D and I are declination and inclination of the mean direction; k and α_{95} are Fisher's precision parameter and radius of the 95% circle of confidence ($P = 0.05$); dp and dm are the semi-angles of the oval of confidence about the pole position.

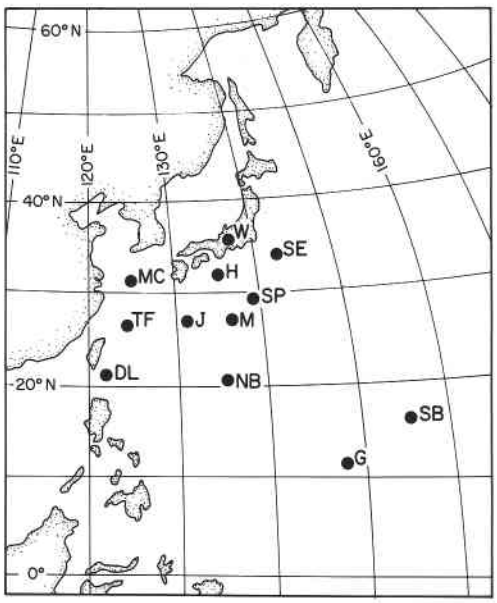


FIG. 11. Paleopoles of selected rock units considered representative of the Carboniferous paleomagnetic field relative to North America. DL, Deer Lake Group (Visean to early Namurian) (Irving and Strong 1985); G, Gaspé dykes (Early Carboniferous (Seguin 1987); H, Horton Group (Tournaisian) (Scotese *et al.* 1984); J, Jeffrey's Village Member (Visean) (Murthy 1985); MC, Mauch Chunk Formation (Mississippian) (Kent and Opdyke 1985); NB, New Brunswick volcanics (late Namurian – early Westphalian) (Seguin *et al.* 1985); SB, Sandwich Bay dykes (327 Ma) (this study); SE, Searston Formation (Namurian–Westphalian) (Murthy 1985); SP, Spout Falls Formation (Tournaisian) (Murthy 1985); TF, Terrenceville Formation (Tournaisian) (Kent 1982); W, Windsor Group (Visean) (Scotese *et al.* 1984). Mean of all Carboniferous poles listed above is shown as M.

Some alternatives may be considered for the location of the Sandwich Bay (SB) pole vis-à-vis that of other Carboniferous poles. It is possible that the radiometric age cited in Table 1 may be too young. The SB pole, however, is not too far from the Gaspé pole (Seguin 1987), for which there is good age control. The Gaspé dykes cut Devonian sedimentary units, and they are certainly younger than 370 Ma. The SB pole is a primary pole and thus is probably representative of the Early to Middle Carboniferous magnetic field. It is interesting to note that the Sandwich Bay dykes and the Gaspé dykes both have a

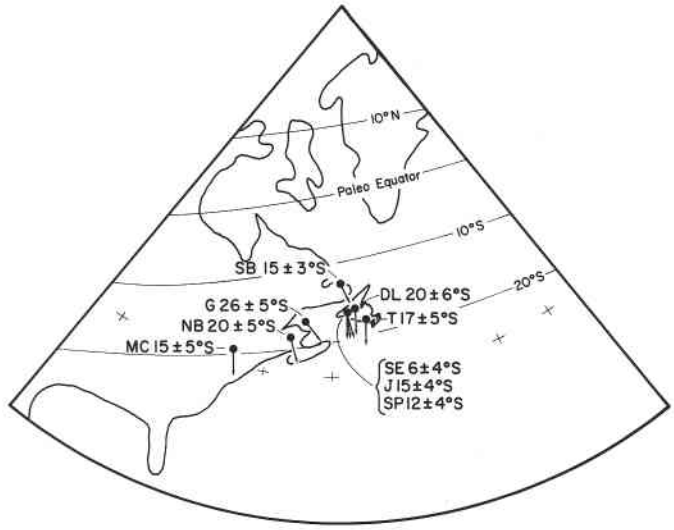


FIG. 12. Paleolatitudes of North America during the Carboniferous, as estimated from paleopoles listed in Fig. 11.

similar east-southeast trend and occur in a similar tectonic position, viz. in the stable craton on the northwest side of the Carboniferous basins.

The SB, Gaspé (G), and to a lesser extent the New Brunswick (NB) poles have distinctly lower latitudes than the other Carboniferous poles. It is to be noted that the rest of the Carboniferous poles are all from sedimentary units. Further, only SB, G, and NB poles (all for igneous rocks) do not have any Kiaman overprints. A more important parameter than the paleopole would be the paleolatitudes of the site and of North America during the Carboniferous times. These were calculated from the results and plotted in Fig. 12. The paleolatitudes are in excellent agreement for various rock units, and the data clearly show that relevant parts of North America were between 15° and 20° south of the paleoequator during mid-Carboniferous times. There are slight differences in declination for the various rock units as depicted in Fig. 12. This suggests there may be the possibility of small relative rotations between sampling areas about a vertical axis.

Conclusions

(1) A suite of Carboniferous dykes from the Sandwich Bay area of coastal Labrador is recognized as distinct from other dykes in the area based on their geochemistry–field characteristics and their paleomagnetic behaviour.

(2) The dykes are chemically homogeneous and are considered within-plate tholeiites.

(3) A K–Ar whole-rock age of 327 ± 13 Ma was determined for a member of this suite of dykes and was interpreted as the age of intrusion of the Sandwich Bay dykes.

(4) Paleomagnetic signature of the dykes is uncomplicated, constituting a PEF component superposed on a characteristic magnetization that is northwest and upward directed. None of the dykes has a Kiaman overprint. Baked-contact tests for various dykes in the group are positive, suggesting that the isolated stable remanence of the dykes is primary.

(5) The paleopole position for this suite of dykes is at 16.2°N, 157.6°E, which is interpreted as representative of the paleomagnetic field 327 Ma ago. The corresponding paleolati-

tude is at 15°S ± 3°. It is suggested, based on this result, that North America was south of the paleoequator by between 15° and 20° during Early to Middle Carboniferous times.

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